Nyaya Saathi: AI legal first-aid for India (Problem 2)

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1. Why this problem, and what I assumed

A confident wrong answer about law can cost someone their deposit or their deadline, so **refusal design and checkable output** are the product, not fluent text. I built for a first-time, non-lawyer user typing English, Hindi or Hinglish. **Assumptions (brief was silent):**

• **Scope = six everyday problems:** security deposit, consumer refund, unpaid salary, bounced cheque, online/UPI fraud, notice received. Family, criminal defence, property/builder, inheritance, tax, insurance, PF, job termination, harassment and loans between friends are referred, not answered.

• **Law as of 8 Nov 2026** from a hand-curated KB of 18 entries, each with a source. Three entries were re-checked on the web on 8 Nov (labour codes in force from 21 Nov 2025 and the old Wages and Gratuity Acts repealed; e-Jagriti replaced e-Daakhil on 1 Jan 2025; nil fee up to Rs 5 lakh). The other entries are from memory and carry a 'verify' flag where unsure; a lawyer must review them before real use.

• First-aid only: understand, prepare, route. No representation, no prediction of who wins. Text only; voice would be a speech-to-text layer in front (not built).

2. How it works: what is code, what is AI, where a human sits

Per message: (1) **Triage in code**: language, category (whole-word keywords), red flags (arrest, violence, blackmail, self-harm, eviction, criminal threat, user is on the other side), amount, limitation clue, and **deadline arithmetic** for cheques (30 days from the return memo, computed from a date or '40 days ago'). (2) **Retrieval in code**: the KB entries for that category; the model never chooses the law. (3) **LLM only phrases** the answer from that context with [TAGS], and extracts a case file (what/when/proof). (4) **Guard in code** deletes unsupported sentences (section 3). (5) **Templates** give a notice draft with [BRACKETS] for unknown facts, plus a role-play prompt (AI plays landlord/HR/seller). Self-harm returns a fixed crisis message only, with no case advice. Out-of-scope and red-flag cases put 'see a lawyer / NALSA 15100' first and suppress the notice. The model is one function (`call_llm`, env vars) so Gemini, Claude or any OpenAI-compatible API can be swapped; on any error it falls back to offline mode. **Human:** a lawyer reviews KB entries and sampled chats.

3. How it avoids inventing law

The prompt is closed-world (CONTEXT only), then a guard removes any sentence that fails one of six checks: unknown Act (with or without a year; wrong year caught), section/article/rule numbers and Section-to-Act pairing, bare abbreviations (IPC, BNS, CrPC, NI Act), courts and case names, Hindi statute words not mapped to a KB Act, and **duration figures** ('7 days', 'six months') not found in the KB or the user's own words. **Limit:** a wrong fact with no number or name (e.g. 'the owner must be reached first') passes, and a figure the user mentioned can be repeated back wrongly. So KB review is the real control and the guard is the safety net.

4. Scale and cost

Offline mode is free. LLM mode is 2 calls per turn, about 3k tokens in and 1k out (my estimate, not measured), a small fraction of a rupee on a small model; verify with real usage. To scale: per-state KB records with owner and 'last verified' date; embedding retrieval; lawyer review of new entries and a weekly 2% chat sample; the test files run as a release gate on every KB or model change. A new problem type is one dictionary entry plus 3–5 tests.

5. How I know it works, and where it fails

My first version passed its own 20 tests, which proved little because I wrote both. So I audited it with inputs it had never seen:

• **Guard audit:** 15 adversarial sentences (bare 'IPC 420', 'u/s 420', Hindi Act name, 'Rent Control Act', 'Article 21', a court case without 'v.', 'within 7 days by law', 'six months' rent'). The first guard caught **3 of 15**. After the rewrite: 18/18 adversarial caught and 6/6 legitimate sentences kept (24-case suite, plus a realistic mock model answer lost 0 sentences).

• **30 fresh messy inputs** found: PF dispute routed to the deposit flow ('depositing' contains 'deposit'); '4 years 8 months' read as **Rs 8** ('years 8'); no cheque deadline maths at all (a memo 45 days old got no warning); a self-harm message got the full salary advice and 'do you have an offer letter?'; blackmail not escalated; a landlord chasing unpaid rent got tenant advice; a legal threat labelled 'violence'; the violence and out-of-scope path lacked the 181/112 helplines; **labour law presented as pre-2025**.

• **Held-out round** (12 new inputs, written before fixing): about 6 handled well, 3 failed safe ('cannot tell, tell me more'), **3 unsafe** (two criminal threats not escalated, a rent increase given deposit advice). Fixed and added as tests.

• Now: 43/43 triage cases and 25/25 guard probes pass offline (raw files in outputs/; run date pinned for deadline tests). **Caveat:** the later cases were written after seeing failures, so this is regression safety, not a generalisation score.

Still fails: keyword triage misses new phrasings (PG 'advance', 'last month's pay' fall to 'cannot tell', which fails safe) and picks one problem when several are mixed; a bare '20000' with no currency word is not read as an amount; offline Hindi covers headings only; the chat re-analyses the user's messages but the model never sees its own earlier replies; notice templates are not state-specific; no voice. **LLM mode:** implemented and swappable (Gemini / Claude / OpenAI-compatible), but the 43 cases in outputs/ were run in offline mode; a full LLM-mode run is not yet recorded.

6. Tools used, and where they got it wrong

Claude (design, code, KB drafting), a second pass of Claude plus web search for the audit and fact-checks, Streamlit for the UI. **Where the AI got it wrong:** it wrote the guard and the tests together, then reported 20/20 and a 'four-layer' defence; my audit showed the guard missed most evasions, the self-harm claim in the draft doc was not what the code did, and its KB cited the repealed Payment of Wages Act and Gratuity Act as live law.